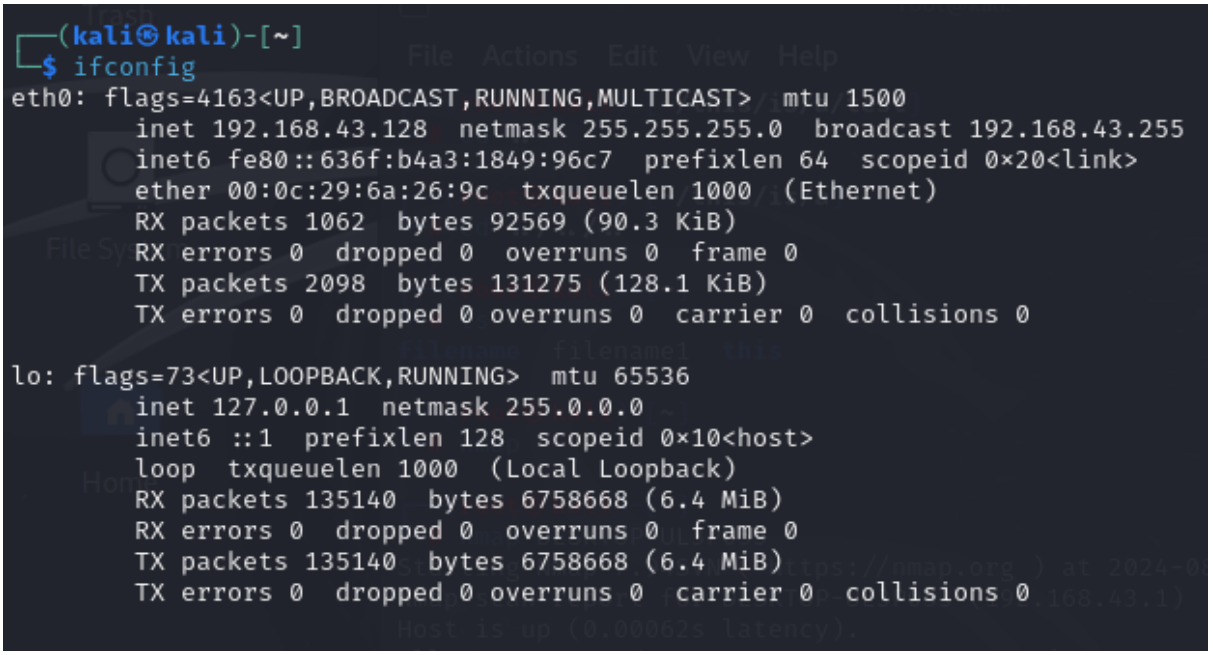
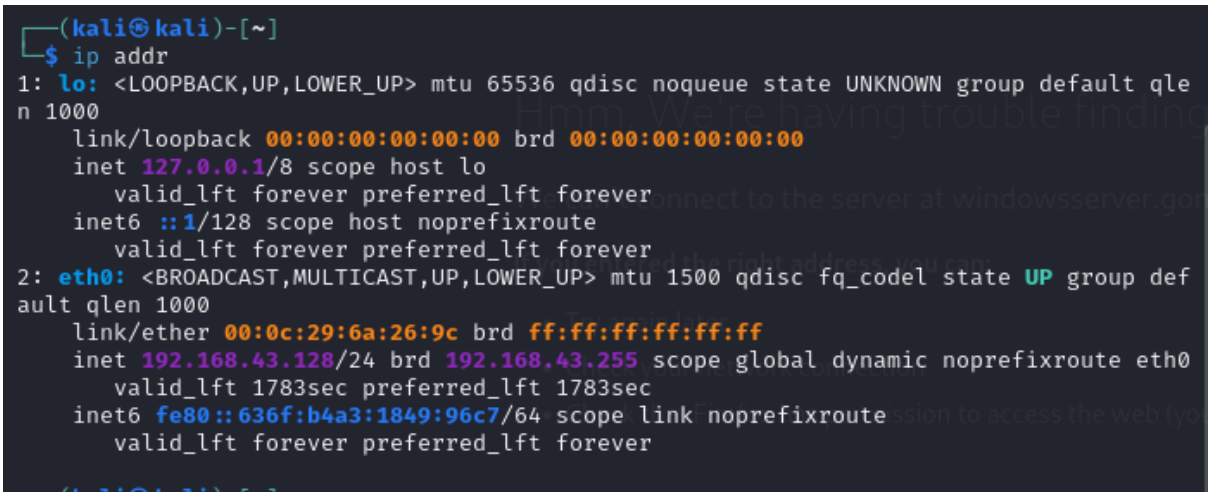
NAME: ILORI BLESSING DAVID
COHORT: SUNGLOW MAULERS
CHECKPOINT: NETWORKING

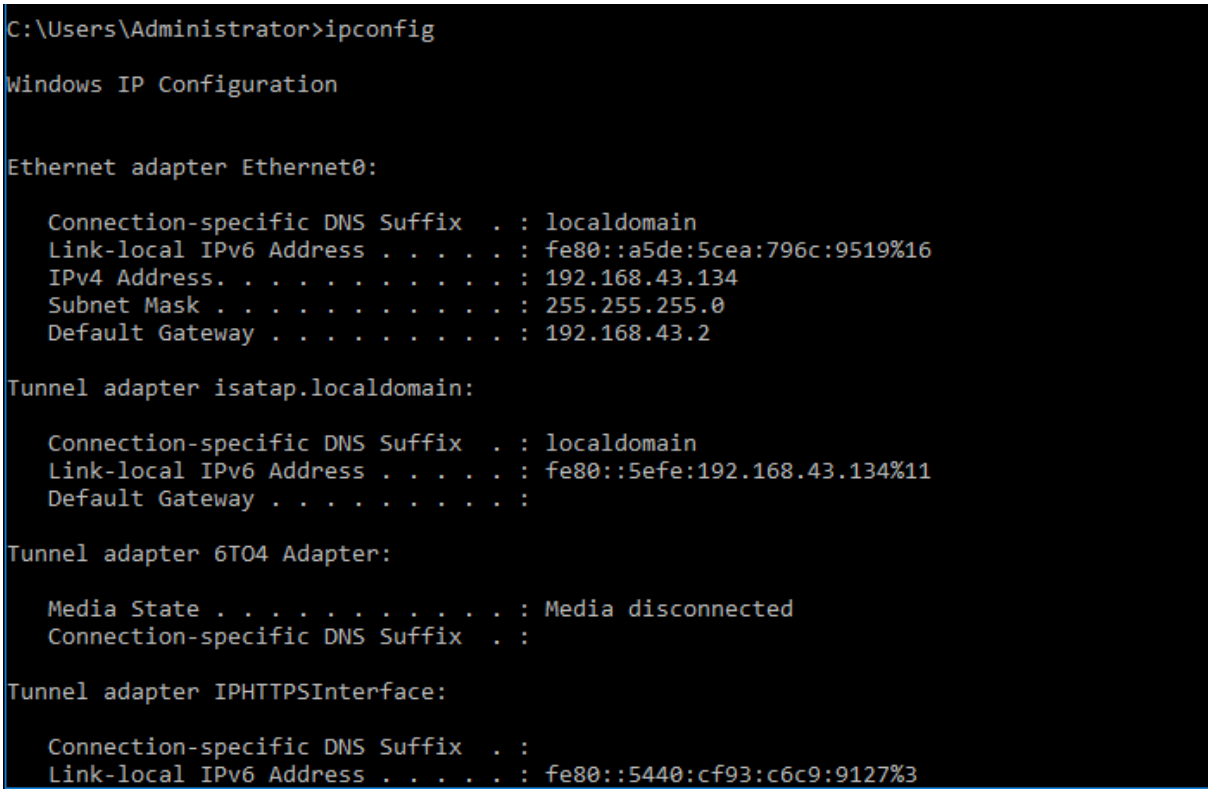
In this checkpoint, I will describe how I conducted network reconnaissance by identifying various nodes through the use of different tools.

I'm looking to identify IP addresses across different operating systems—Kali Linux, Windows, and Debian. Each OS has its own tools for this task. For example, ifconfig and ip addr work on Kali Linux, as shown in the image below while the command was being executed.



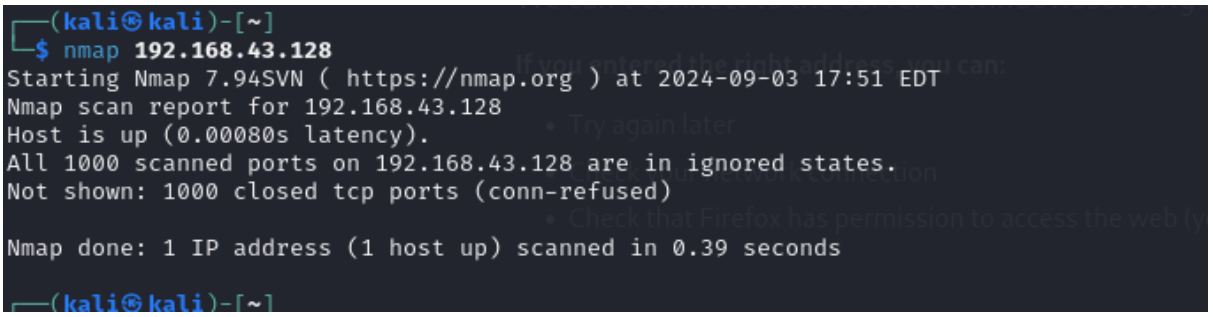


Continuing with identifying IP addresses, the ipconfig command works on Windows and displays the IP address, as shown in the image below.



DEBIAN YET TO INSTALL.

After identifying IP addresses, I plan to use Nmap, a tool for network discovery, port scanning, OS detection, vulnerability scanning, security auditing, and network performance testing. For this checkpoint, I will focus on using Nmap to identify open ports on various machines. I did a basic scan of ***nmap 192.168.43.128***, see image below



I did a range scan also using **nmap -p- for all 65535 ports** see image below

```
┌──(kali㉿kali)-[~]
└─$ nmap -p- 192.168.43.128
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-09-02 09:15 EDT
Nmap scan report for 192.168.43.128
Host is up (0.00041s latency).
All 65535 scanned ports on 192.168.43.128 are in ignored states.
Not shown: 65535 closed tcp ports (conn-refused)

Nmap done: 1 IP address (1 host up) scanned in 15.99 seconds
```

Using nslookup, a command-line tool for obtaining domain name or IP address mapping information, I want to identify the Fully Qualified Domain Name (FQDN) of the IP address ***nslookup 3.33.130.190***. See the image below for details

```
┌──(kali㉿kali)-[~]
└─$ nslookup 3.33.130.190
190.130.33.3.in-addr.arpa       name = a2aa9ff50de748dbe.awsglobalaccelerator.com.

Authoritative answers can be found from:
```

Now for nslookup of of the domain name

```
┌──(kali㉿kali)-[~]
└─$ nslookup awsglobalaccelerator.com
Server:         192.168.43.2
Address:        192.168.43.2#53

Non-authoritative answer:
*** Can't find awsglobalaccelerator.com: No answer
```

The ping command is used to test the reachability of a networked device and measure the round-trip time for messages sent from the originating host to the destination device. Therefore, I will use ping to verify communication with the IP address ***ping 172.16.103.134.***

```
┌──(kali㉿kali)-[~]
└─$ ping 172.16.103.134
PING 172.16.103.134 (172.16.103.134) 56(84) bytes of data.
^C
── 172.16.103.134 ping statistics ──
17 packets transmitted, 0 received, 100% packet loss, time 16365ms
```

The dig command provides detailed information about domain names, IP addresses, and DNS records. To use dig to identify the Fully Qualified Domain Name (FQDN) of the IP address

associated by looking into the domain name as shown in the image below ***dig 172.16.103.134***

```
┌──(kali㉿kali)-[~]
└─$ dig 172.16.103.134

; <<>> DiG 9.20.0-Debian <<>> 172.16.103.134
;; global options: +cmd
;; Got answer:
;; →→HEADER←← opcode: QUERY, status: NXDOMAIN, id: 21178
;; flags: qr rd ra; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; MBZ: 0×0005, udp: 4096
;; QUESTION SECTION:
;172.16.103.134.                        IN      A

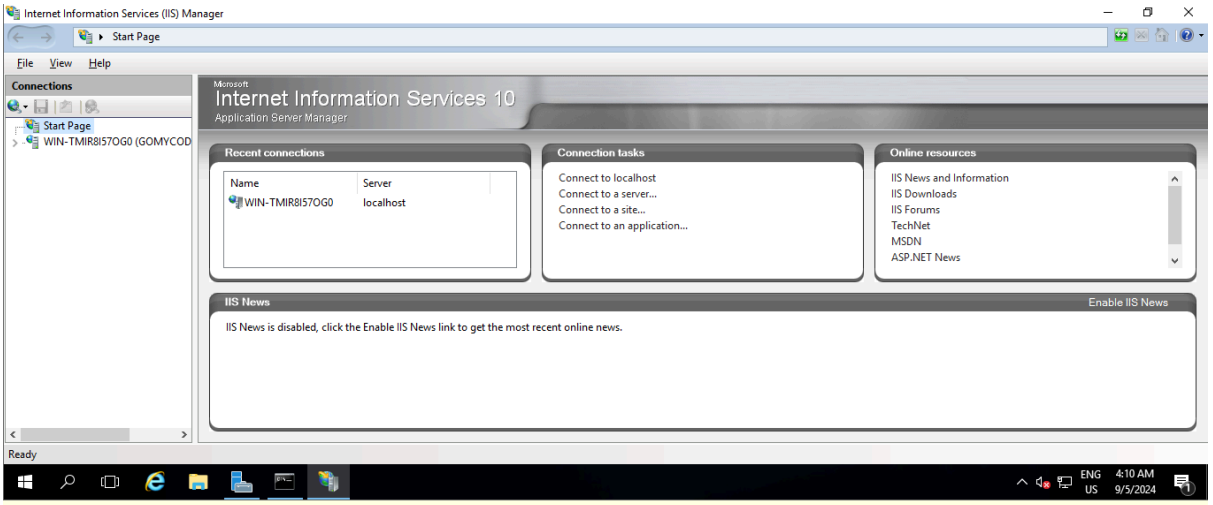
;; AUTHORITY SECTION:
.                       5       IN      SOA     a.root-servers.net. nstld.ver
isign-grs.com. 2024090602 1800 900 604800 86400

;; Query time: 706 msec
;; SERVER: 192.168.43.2#53(192.168.43.2) (UDP)
;; WHEN: Fri Sep 06 18:14:20 EDT 2024
;; MSG SIZE  rcvd: 118
```

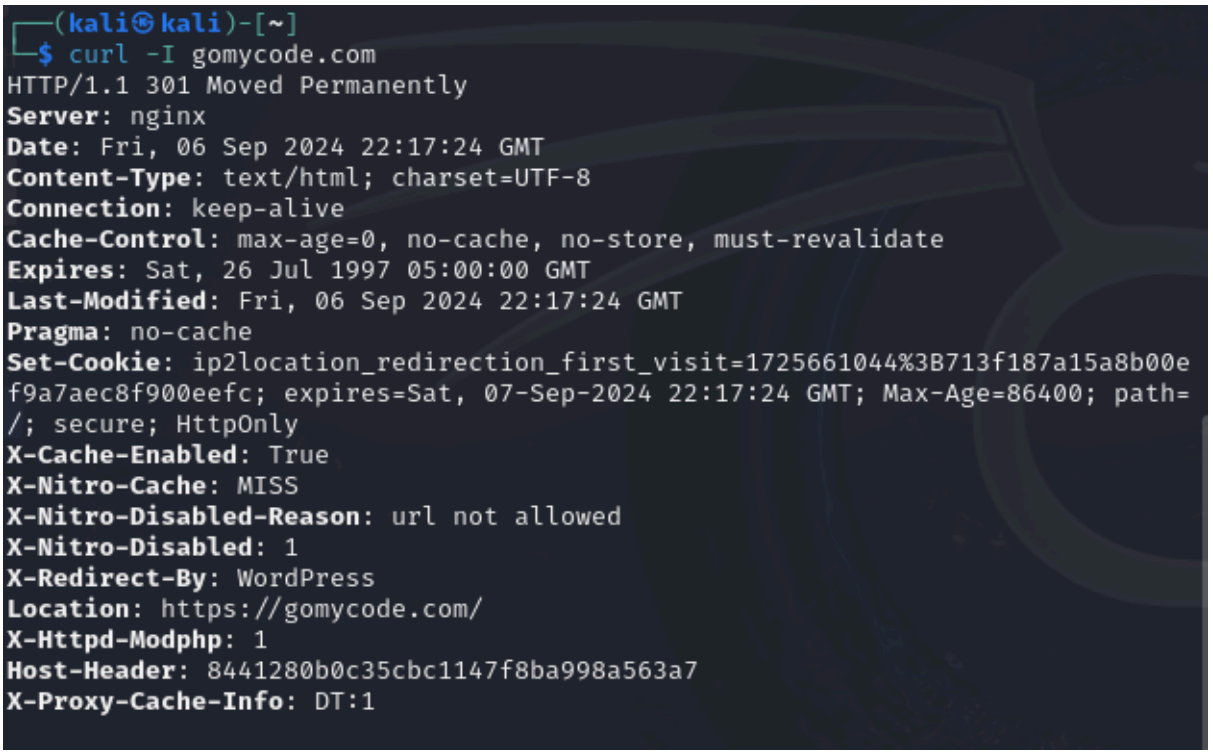
LETs re verify by pinging ***ping gomycode.com***

```
┌──(kali㉿kali)-[~]
└─$ ping GoMyCode.com
PING GoMyCode.com.localdomain (35.207.134.165) 56(84) bytes of data.
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=1
 ttl=128 time=165 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=2
 ttl=128 time=168 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=3
 ttl=128 time=147 ms
From 192.168.43.128 icmp_seq=5 Destination Host Unreachable
From 192.168.43.128 icmp_seq=6 Destination Host Unreachable
From 192.168.43.128 icmp_seq=7 Destination Host Unreachable
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=4
 ttl=128 time=5032 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=8
 ttl=128 time=159 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=9
 ttl=128 time=176 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=1
0 ttl=128 time=151 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=1
1 ttl=128 time=172 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=1
2 ttl=128 time=2031 ms
64 bytes from 165.134.207.35.bc.googleusercontent.com (35.207.134.165): icmp_seq=1
3 ttl=128 time=1012 ms
```

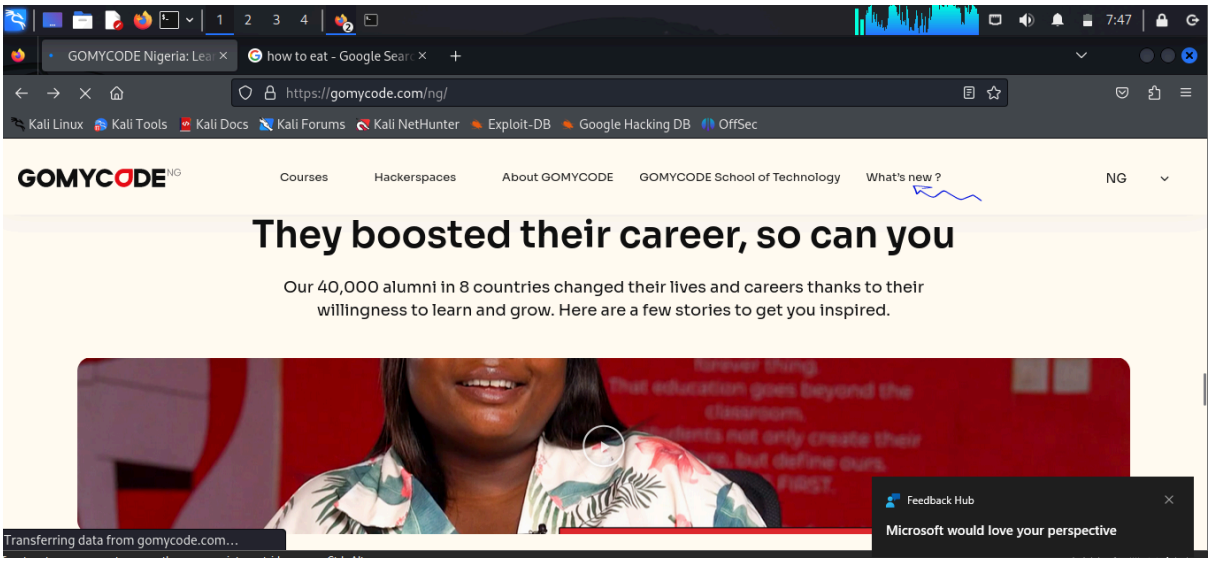
Windows IIS RUNNING see image below;



Using curl let's identify the server running on a url ***curl -I gomycode.com.***



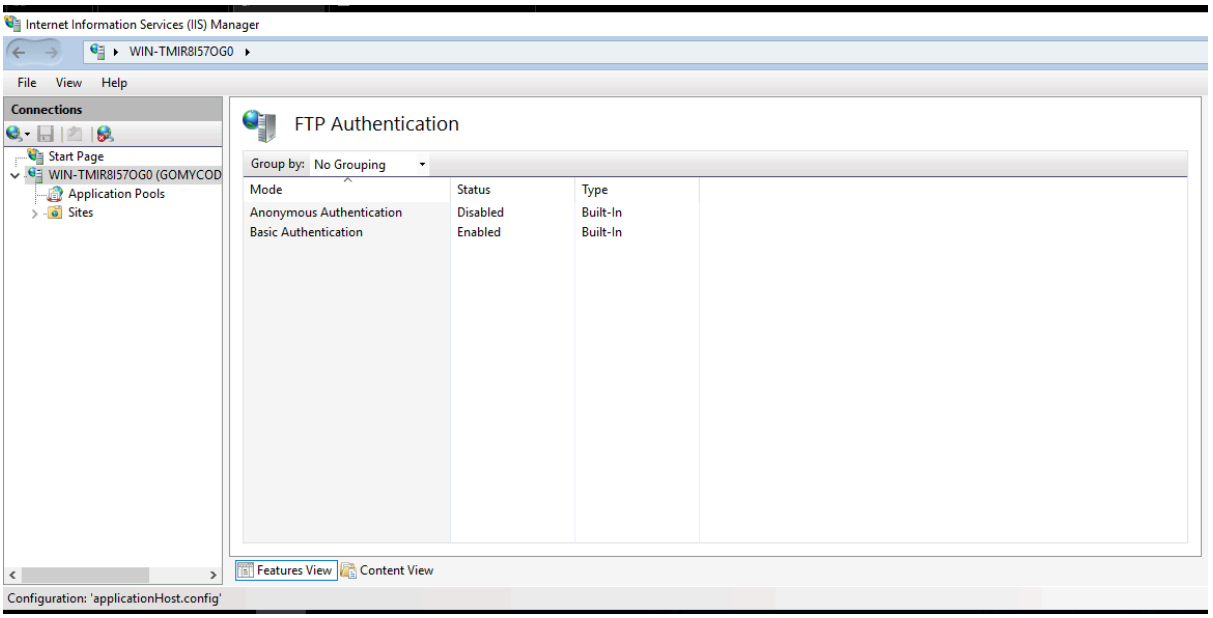
YES i can access the windows website www.gomycode.com on kali linux.. See image below



NEXT:

On the windows server go to **tools -> internet information service (IIS).** click on the server name on the IIS section under ftp section click on **: authentication.**

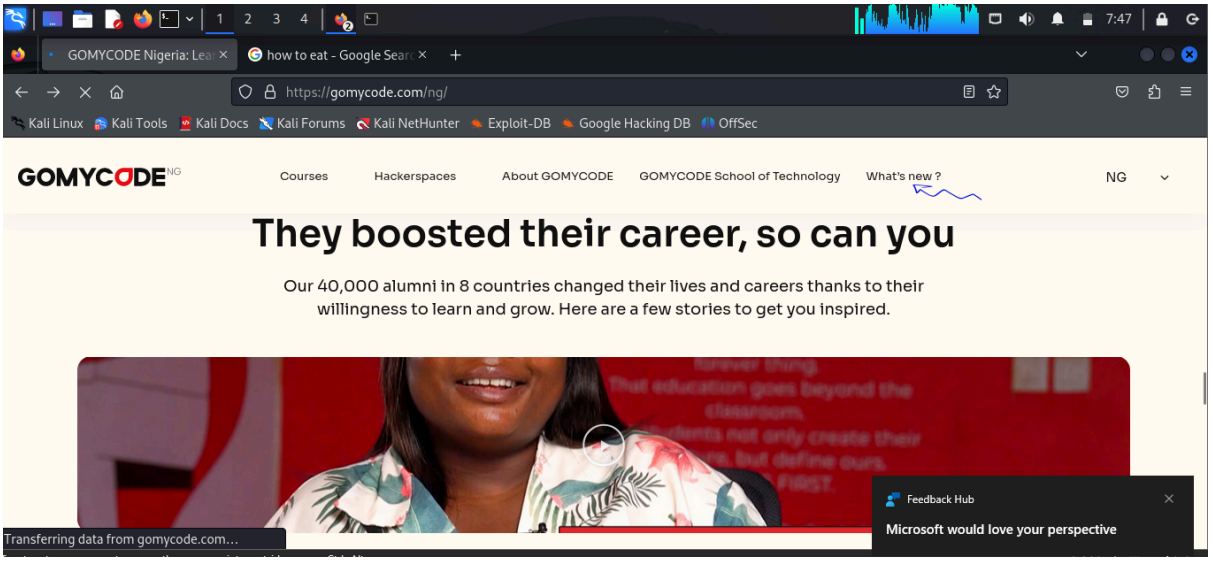
- **right click on Anonymous Authentication and choose disable**
- **right click on the Basic Authentication and choose enable**



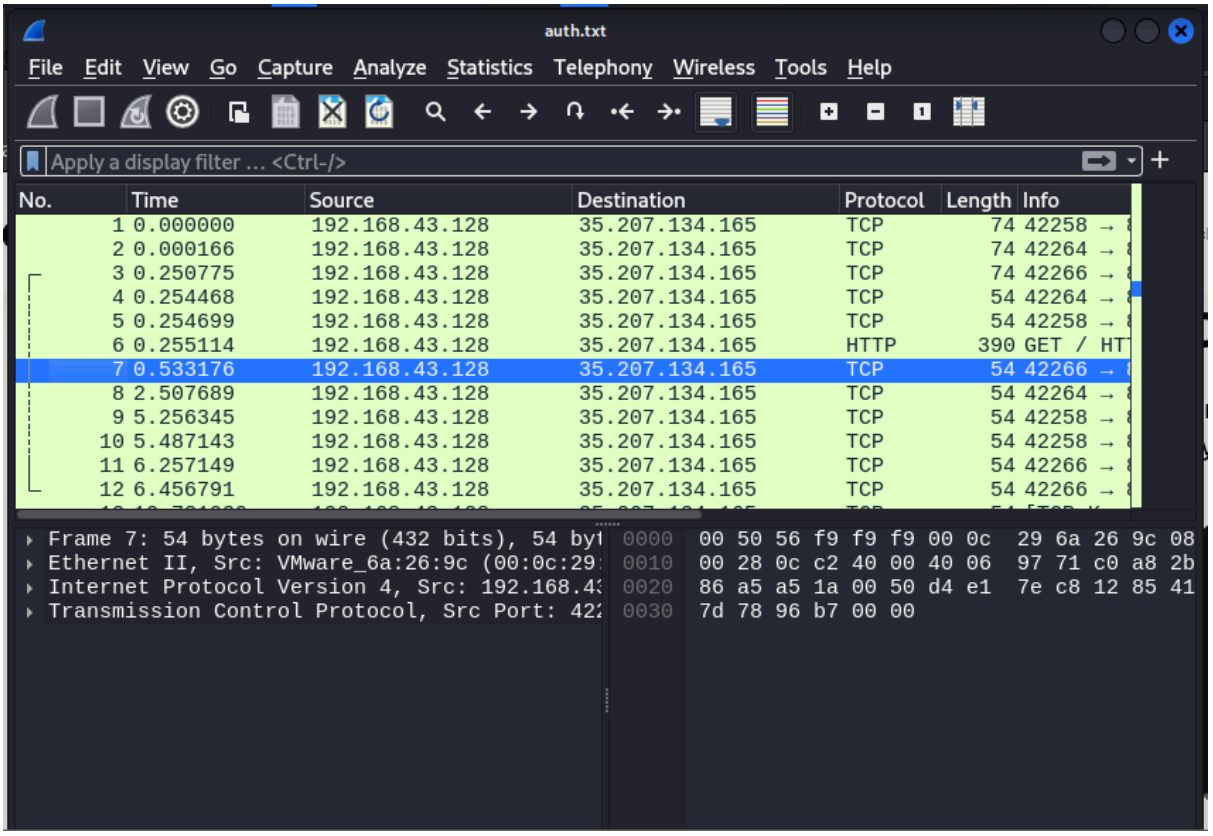
So running this commands to sniff the network
Using the tcpdump command which is used to analyse network traffic by intercepting packets that are being received by the computer running on.
So running command ***sudo tcpdump -vv dst gomycode.com and port www -w auth.txt***

```
┌──(kali㉿kali)-[~]
└─$ sudo tcpdump -vv dst www.gomycode.com and port www -w auth.txt
[sudo] password for kali:
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 2621
44 bytes
^C0 packets captured
7 packets received by filter
0 packets dropped by kernel
```

Image of running the site for tcpdump



Launching wireshark to capture packets from a network connection. See image below for ***wireshark auth.txt***



THANK YOU.